Question 1

Find the fastest safe clock frequency for a 3-cycle pipeline that feeds back on itself. Solve using D-FFs for synchronization. Verify that hold time is met.

- $T_{skew} = 0\,\text{ps}$
- 7 stages of combinational logic, each with $T_{cd-group} = 100\,\text{ps}$ and, used in the provided sequence, $T_{pd} = \{500, 600, 300, 400, 500, 200, 110\}$ ps. Synchronize the 110 ps stage output to the clock.
- latches and D-FFs : $T_{setup} = 40\,\text{ps}$, $T_{hold} = 30\,\text{ps}$, $T_{pcq} = 45\,\text{ps}$, $T_{ccq} = 30\,\text{ps}$, $T_{pdq} = 50\,\text{ps}$, $T_{cdq} = 35\,\text{ps}$

We are going to start off with our assumptions. We assume here that the combinational component must be executed in order, and the order is given by listing in the question. This means that you could only group stages adjacent to each other. Additionally, there shall be 3 stages of combinational logic. With these assumptions, we can now start to build our pipeline.

In our vision, the new pipeline shall have the following grouping of combinational circuits.

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{500 + 600, 300 + 400, 500 + 200 + 100}
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Where the first and second group has two stages, and the third group has 3 stages. The system is available in figure 1.

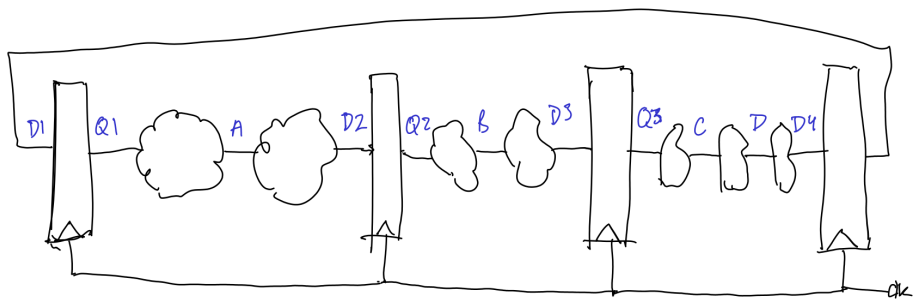

Figure 1: The block diagram for the proposed system.

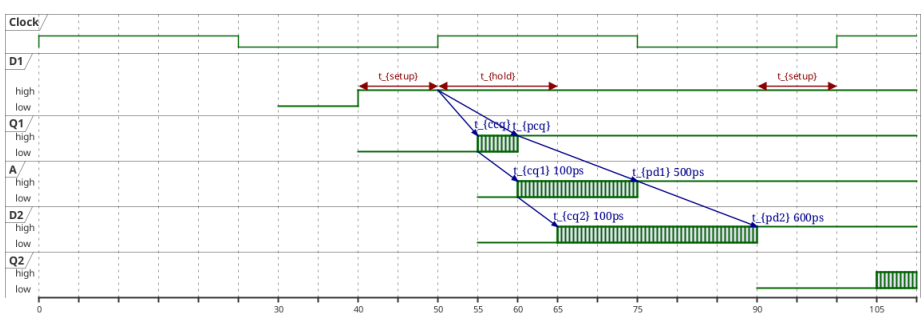

Figure 2: Timing diagram used to understand how the configuration of the system affects the timing parameters.

$$T_c \geq t_{pcq} + t_{pd} + t_{setup} + t_{skew} \tag{1}$$

$$t_{hold} + t_{skew} \leq t_{ccq} + t_{cd} \tag{2}$$

For a working pipeline architecture, the minimum period is the time it takes to correctly execute the slowest stage of the pipeline. We have made the first stage of the pipeline the slowest stage of the pipeline. From here, we do our computation in a spreadsheet, illustrated in table 1. The maximum frequency with no skew was found to be 840.336 MHz.

Table 1: The computed values. The value of minimum T_c was computed from (1). The verification of the hold time is done through (2). In this case, $t_{skew} = 0$. In this case $t_{hold} = 30$ ps. The frequency was found to be 840.336 MHz.

Stage	t_{pd}	total contamination delay	min T_c	meets hold time
1	1100	200	1190	yes
2	700	200	790	yes
3	800	300	890	yes
Max period	1190			
Max frequency	840336134.4537815			

Question 2

Repeat with $T_{skew} = 50$ ps, direction unknown.

With unknown direction, we should put the skew in the worse case direction for both the hold time and setup time constraints. With this, we could use (1) and (2) to find out what the safest safe clock frequency is, and if hold time is violated. The table with the intermediate results of the computation is available in table 2. The maximum frequency with the skew at 50 ps was found to be 806.45 MHz.

Table 2: The computed values. The value of minimum T_c was computed from (1). The verification of the hold time is done through (2). In this case, $t_{skew} = 50$ ps. Additionally, the effective hold time with the skew applied is $t_{hold} = 80$ ps. The frequency was found to be 806.45 MHz.

Stage	t_{pd}	total contamination delay	min T_c	meets hold time
1	1100	200	1240	yes
2	700	200	840	yes
3	800	300	940	yes
Max period	1240			
Max frequency	806451612.9032258			